

SYSTEM FOUNDRY - PERSONAL MVP

SpoolReserve

Know whether the spool can finish before the print begins.

Project	2026-08-15-0901-spoolreserve
Customer	Hobby FDM users and small maker studios
Price	One-time KRW 9,900
Status	PASS - automated tests 10/10

1. Project overview

SpoolReserve is a local-first web tool that converts measured gross and empty-spool weights into real filament remaining, then checks whether that spool can finish a selected print job.

2. Target customer

Hobby FDM users, school maker clubs and small print studios that keep several partially used filament spools.

3. Problem

Slicer estimates do not reveal whether an opened spool has enough material. Guessing can waste machine time and leave unfinished jobs. The tool adds a user-selected safety margin before matching inventory.

4. Core features

- 1) Validated spool registration from measured weights.
- 2) Material-aware job feasibility with safety margin.
- 3) Consumption and remaining stock-value tracking.

5. Usage

Run the static server, register each spool, then enter the slicer estimate, material and safety percentage. Select a fitting spool and record the estimated consumption after a successful print.

6. System structure

Browser UI -> validation and calculation engine -> localStorage. No account, server database or paid API is required.

7. Technology

HTML5, CSS, modern JavaScript modules, Node.js built-in test runner, Python static server and Docker.

8. Running screen

KNOW BEFORE YOU PRINT.

LOCAL DATA - NO ACCOUNT

SpoolReserve

Match real filament weight to the next job, with safety margin included.

Active spools

3

Filament left

1691g

Stock value

KRW 42,715

Job matches

2

Add a spool

Spool code

PLA-GRY-03

Material / color

PLA · Graphite

Gross / empty weight

910g / 245g

Purchased net / price

1000g / KRW 24,000

SAVE SPOOL

Measured weight minus empty spool tare.

JOB FEASIBILITY

Choose the spool that can finish

202g incl. margin

Job

Material

Estimate

Safety

Enclosure set

PLA

180g

12%

CHECK FIT

PLA

Matte Black · PLA-BLK-01

645g left · KRW 14,190 stock value

3 complete runs

202g needed

PLA

Pearl White · PLA-WHT-02

218g left · KRW 5,341 stock value

1 complete runs

202g needed

PETG

Clear · PETG-CLR-01

828g left · KRW 23,184 stock value

Other material

202g needed

Rendered from the same calculation engine verified by automated tests.

Application state rendered from the verified production calculation engine.

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9. Test result

Automated tests 10/10 passed. They cover normalization, invalid weights, remaining grams, proportional value, safety rounding, fit and material mismatch, queue order, duplicate IDs and consumption. HTTP start, README commands and a full calculation workflow were also verified.

10. Security and limitations

Inputs are length-, range- and allow-list validated; UI output is escaped. Data remains local. The estimate does not account for failed prints, purge towers, moisture, inaccurate scales or material left in the feed path. A practical margin and physical verification remain necessary.

11. Price and revenue model

One-time KRW 9,900 purchase. The pitch is avoiding one failed long print or one prematurely replaced spool.

12. First-customer plan

Offer a free three-spool inventory setup to ten local maker-club and 3D-printing community users. Show grams and stock value recovered, then convert useful pilots to the paid copy.

13. Improvement plan

1) Reusable empty-spool tare library. 2) QR labels. 3) Scale import. 4) Purge and support presets. 5) CSV backup. 6) Multi-device sync.

14. Overall evaluation

Evaluation	Score	Reason
Problem intensity	8/10	A wrong estimate wastes time and material.
One-day completeness	9/10	Registration, matching and consumption are complete.
Sales potential	8/10	Value maps directly to avoided failed prints.
Maintenance ease	9/10	Static app with no external service.
Differentiation	8/10	Measured tare, safety and stock value share one flow.
Expansion potential	8/10	Scale, QR and multi-device features fit naturally.

Overall score 8.4/10

A narrow, immediately understandable workflow with measurable saved material. Before paid release, validate tare-library and scale-import priorities with real users.

Appendix - verification checklist

PASS: zero external dependencies; program start; core workflow; 10 automated tests; README commands; credential scan; PDF parsing and all-page rendering.

Release decision

PASS - suitable for local use and customer interviews. Users must still verify physical spool and printer conditions before long jobs.